

Math 421

Wednesday, October 8

Announcements

- Homework 5 due Friday
- Grace's Drop-In Hours
 - **Today:** 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Exam 1 – 10/15
 - If you have a conflict, email me (if you haven't already) and **fill out makeup exam request form** (linked from Exam Information page on Canvas) ASAP!
 - If you are a McBurney student, sign up to take the exam at the testing center through RegisterBlast.

Algebraic Limit Laws

Theorems 9.2-9.6: Let $\lim s_n = s$ and $\lim t_n = t$. Then,

i. $\lim ks_n = ks$ for all $k \in \mathbb{R}$.

ii. $\lim s_n + t_n = s + t$.

iii. $\lim s_n t_n = st$.

iv. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{1}{s_n} = \frac{1}{s}$

v. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$

FALSE Proof of iv. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{1}{s_n} = \frac{1}{s}$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since (s_n) is convergent, there is a constant $M > 0$ such that $|s_n| \leq M$ for all n . Since $\lim s_n = s$, there exists $N > 0$ such that if $n > N$, then $|s_n - s| < \epsilon M |s|$. Assume $n > N$. Then

$$\left| \frac{1}{s_n} - \frac{1}{s} \right| = \left| \frac{s - s_n}{s_n s} \right| \leq \frac{|s_n - s|}{M |s|} < \frac{\epsilon M |s|}{M |s|} = \epsilon$$

Since $\epsilon > 0$ was arbitrary, $\lim \frac{1}{s_n} = \frac{1}{s}$.

Proof of iv.

Lemma 9.5: Let $\lim s_n = s$, if $s_n \neq 0$, for all n , and if $s \neq 0$, then

$$\lim \frac{1}{s_n} = \frac{1}{s}.$$

Discussion.

Proof of iv.

Lemma 9.5: Let $\lim s_n = s$, if $s_n \neq 0$, for all n , and if $s \neq 0$, then

$$\lim \frac{1}{s_n} = \frac{1}{s}.$$

Proof.

FALSE Proof of v. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since (s_n) is convergent, $s_n \neq 0$ for all n , and $\lim s_n = s \neq 0$, there is a constant $m > 0$ such that $|s_n| \geq m$ for all n . Since $\lim s_n = s$, there exists $N_1 > 0$ such that if $n > N_1$, then $|s_n - s| < \frac{\epsilon|s|m}{2|t|}$. Since $\lim t_n = t$, there exists $N_2 > 0$ such that if $n > N_2$, then $|t_n - t| < \frac{\epsilon M}{2}$. Let $N = \max\{N_1, N_2\}$.

Assume $n > N$. Then

$$\begin{aligned} \left| \frac{t_n}{s_n} - \frac{t}{s} \right| &= \left| \frac{t_n s - t s_n}{s_n s t} \right| = \left| \frac{t_n s - t s + t s - t s_n}{s_n s} \right| \leq \left| \frac{t_n s - t s}{s_n s} \right| + \left| \frac{t s - t s_n}{s_n s} \right| \\ &= \left| \frac{t_n - t}{s_n} \right| + \left| \frac{t(s - s_n)}{s_n s} \right| < \frac{1}{m} \frac{\epsilon m}{2} + \frac{|t|}{m|s|} \frac{\epsilon|s|m}{2|t|} = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim \frac{s_n}{t_n} = \frac{s}{t}$.

Proof of v.

Theorem 9.6: Let $\lim s_n = s$ and $\lim t_n = t$. Then, if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$.

Proof.

Example: $\lim \frac{4n^3+3n}{n^3-6} = 4$

Activity – Critique the Proof

Claim: $\lim_{n \rightarrow \infty} \frac{1}{n^{1/4}} = 0$.

FALSE Proof. We know that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. So, by our algebraic limit theorems,

$$\left(\lim_{n \rightarrow \infty} \frac{1}{n^{1/4}} \right)^4 = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

So $\lim_{n \rightarrow \infty} \frac{1}{n^{1/4}} = 0$.

9.7 Theorem (Basic Examples)

- a) $\lim \left(\frac{1}{n^p} \right) = 0$ for $p > 0$.
- b) $\lim a^n = 0$ if $|a| < 1$.
- c) $\lim (n^{1/n}) = 1$.
- d) $\lim (a^{1/n}) = 1$ for $a > 0$.

Activity – Prove 9.7 a)

Theorem. $\lim \left(\frac{1}{n^p} \right) = 0$ for $p > 0$.

Discussion.

Proof.

Diverging to $\pm\infty$

Definition 9.8

- $\lim s_n = +\infty$ provided for each $M > 0$ there is a number N such that $n > N$ implies $s_n > M$.
- $\lim s_n = -\infty$ provided for each $M < 0$ there is a number N such that $n > N$ implies $s_n < M$.

Terminology: A sequence (s_n) has a **limit** or the **limit exists** if (s_n) converges to a real number or diverges to $\pm\infty$.

Example: Show that $\lim n^2 = +\infty$

Activity

1. Prove, directly from the definition, that $\lim(e^n - 7) = +\infty$.
2. Prove that the unbounded sequence $s_n = (-1)^n n$ does not have $-\infty$ or $+\infty$ as a limit.
 - a) What does it mean for an unbounded sequence to not diverge to $+\infty$ or $-\infty$, i.e. what is the negation of the definition?
 - b) What proof technique should you use?

Caution!

The limit laws we've previously stated do not apply to sequences that have infinite limits, we need to prove these separately.

Discussion Question: Can you think of some specific examples where the limit laws for finite limits fail if we tried to apply them to infinite limits?